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HAYA A. ALHAKBANI, Al-Imam Mohammad Ibn Saud Islamic University, Riyadh, Riyadh, Saudi Arabia

FATEMA M. ALNASSAR, Goldsmiths, University of London, London, U.K.

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Open Learning Analytics: A Systematic Review of Benchmark Studies using Open University Learning Analytics Dataset (OULAD)

Haya A. Alhakbani*
hahakbani@imamu.edu.sa
hahakbani@imamu.edu.sa
Imam Mohammad Ibn Saud Islamic University (IMSIU)
Riyadh, Saudi Arabia

Fatema M. Alnassar*
ftmalnassar@gmail.com
ftmalnassar@gmail.com
Goldsmiths, University of London
London, UK

ABSTRACT

Virtual learning has gained increased importance because of the recent pandemic situation. A mass shift to virtual means of education delivery has been observed over the past couple of years, forcing the community to develop efficient performance assessment tools. Open University Learning Analytics Dataset (OULAD) is one of the most comprehensive and benchmark datasets in the learning analytics domain. This paper presents the review of benchmark studies performed using OULAD to assess the performance of students in a Virtual Learning Environment (VLE). The presented review aims to highlight the status of technological advancements in this domain and potential future research directions.

CCS CONCEPTS

• **Computing methodologies → Classification and regression trees**.

KEYWORDS

Virtual Learning Environment (VLE), Open Learning Analytics (OLE), Open University Learning Analytics Dataset (OULAD), Students' Performance Evaluation.

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1 INTRODUCTION

In recent years, virtual learning through online platforms (e.g., Open University (OU)) has emerged as a potential means of education mainly because of technological advancements in computers and the easy availability of the internet [8]. The recent COVID-19 pandemic situation has further highlighted the need and importance of virtual education because of the shift in teaching from on-campus to online in most educational institutes [24]. As of now, most educational institutes are using custom online tools and Virtual Learning Environment (VLE) for teaching. Reliable and effective assessment of students' performance on a virtual learning platform is a very challenging and unconventional task to achieve. Unlike traditional on-campus exams, online assessments are vulnerable to cheating and difficult to grade [1, 7]. Each institute is following a different means of assessing students; however, it is far from the real performance of students mainly because of external help, inability to replicate exam room isolated environment and pressure of time. Inability to evaluate the close-to-real performance of students by the OU has raised questions on the quality of education in the past, and on-campus education has always been given preference. However, the recent mass shift to virtual learning due to pandemic has exponentially escalated the issue and forced the community to develop efficient performance assessment mechanisms and tools.

Learning analytics is the domain which deals with the analysis of students' interaction with VLE towards performance assessment. It involves the investigation and analysis of the students/learner's data to improve their online learning experience, mainly by enhancing the guidelines and course material [17]. Machine learning and Artificial Intelligence algorithms have emerged as potential tools that can be used to improve the assessment process given that there is a comprehensive dataset available [22]. In literature, Open University Learning Analytics Dataset (OULAD) [16] released in 2017 is considered one of the most comprehensive OU datasets available. Given the urgency of developing new assessment tools, it is important to review what has already been done using OULAD in the context of students' performance assessment. This article presents a review of benchmark studies performed using the OULAD dataset towards addressing the students' performance on VLE. The aim of the review is to demonstrate the technological trends in this domain and highlight challenges in this context. Furthermore, the presented review intends to indicate the potential future research directions.

2 REVIEW METHODOLOGY

The presented review is based on the standard Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [18] guidelines for systematic reviews. Steps involved in the process included formulating research question/s, listing keywords, deciding academic databases, defining inclusion/exclusion criteria, and analysing selected literature. For this review, "What benchmark

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studies are performed using OULAD to assess the students' performance on VLE?" was explored. A list of corresponding keywords including "OULAD", "VLE", "Student Performance", and "OLA" was formulated to use in combinations for literature search. Academic databases of Web of Science (WoS), IEEE Xplore and Scopus were searched for literature using the formulated keywords. Literature was screened in steps using the inclusion/exclusion criteria. The only journal, conference and book chapter entries were considered in this review written in the English language. Screening steps included duplicates removal, abstract screening, and full-text assessments. The manual full-text assessment was performed based on the question "if literature entry used OULAD dataset in the context of assessing students' performance on VLE?". As a result, 18 entries were selected for final review. Figure 1 shows the PRISMA flow chart for the presented review.

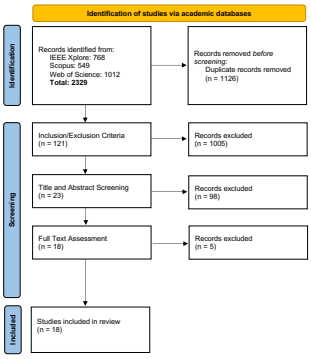

Figure 1: PRISMA Flowchart for Presented Systematic Review (Reconstructed using template provided by Page et al. [21]).

3 OULAD DATASET

OULAD is one of the comprehensive open-source benchmark datasets developed to support research in the learning analytics domain. It has been collected from courses presented at OU and includes both demographic and aggregated clickstream information. In total, data is collected from 22 courses, including 32,593 students. Dataset is anonymous and cannot be linked with the individual student for which the data is collected [16]. OULAD includes three types of data:

(1) Demographics → age, gender region, previous education
(2) Performance → final grades
(3) Behaviour → logs of students' interactions of learning material on VLE.

4 REVIEW OF BENCHMARK STUDIES ON OU DATASET

An author-centred review of benchmark studies performed using the OULAD dataset with the aim to evaluate students' performance on VLE is presented. Review is presented in chronological order to highlight the trends in technological developments in this domain.

Hlosta et al. [13], in the year 2017, proposed a novel ouroboros predictive framework for the identification of at-risk students on VLE. The idea was to use only the data from the ongoing course to train the predictive model instead of using legacy data from previous courses. Based on the data of the currently enrolled course, a predictive model was trained to identify if a student will submit the next assignment or not. Demographic and engagement features from OULAD were used to train the various machine learning predictive models (i.e., Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), Naïve Bayes (NB), XGBoost). From the results, the XGBoost model was reported best. Furthermore, proposed ouroboros was reported better in prediction in the absence of legacy data when compared with traditional legacy data case. Alshabandar et al. [5], in the year 2018, proposed the use of the Gaussian Mixture Model (GMM) for the prediction of early dropout of students in VLE. In addition to GMM. k-Nearest Neighbour (k-NN) and LR models were also trained for performance comparison. Mainly behavioural (i.e., engagement/clickstream) and demographic features of students were extracted from OULAD to train the models. Features were extracted as time-series data for this investigation. The performance of models was assessed based on the F1 score, and from the results, the GMM model achieved the F1 score of 0.835, outperforming k-NN and LR. Although promising results were achieved; however, no comparison with existing literature was performed. In addition, no explanation for the achieved performance of models was presented. In the same year, Heuer and Breiter [10] explored the potential of student's daily activity on VLE to predict performance. Classical machine learning models (i.e., SVM, NB, RF, XGBoost and LR) were implemented on data processed from OULAD. Engagement, performance and demographic features were used to train the models, and the performance of models was assessed using mainly F1 score and accuracy. The importance of binarised activity (i.e., active, non-active) information was reported in the same scope as the importance of exact clickstreams in predicting the students' success. From the results, the SVM model was reported best with an accuracy of 87.98%; however, no comparison with literature was performed. In addition, no discussion on why certain models performed best was presented.

Aljohani et al. [3] in the year 2019 investigated the students' early dropout problem in VLE as a time-series classification problem. Long Short-Term Memory (LSTM) deep learning model was implemented on the clickstream data mined as a time-series data frame from OULAD. The proposed approach outperformed the baseline Artificial Neural Network (ANN) and LR approach by achieving 93.46% prediction accuracy. A weekly breakdown of data was used to study the impact and reported that high prediction could be achieved as early as the first ten weeks. Although the high performance was reported for the proposed approach, however, no comparison with the literature was performed to demonstrate the scope of the presented research. Later in the same year, Jha et al. [14] proposed the use of ensemble learning approaches (i.e., Distributed RF (DRF), Gradient Boosting Machine (GBM), deep learning, Generalized Linear Model (GLM)) for prediction of early dropout students and final result in VLE. Feature engineering was implemented to extract a set of relevant features from OULAD to train the machine learning models. From the results, GBM was reported as best with Area Under Curve (AUC) of 0.91 and 0.93 for dropout and final result prediction, respectively. A brief comparison of literature was

presented to compare the results; however, no explanation for the reported performance was provided.

Chui et al. [6], in the year 2020, proposed a new Reduced Training Vector-Based SVM (RTV-SVM) to predict the at-risk and marginal students in VLE. OULAD was used to train and evaluate the proposed model. From the results, approximately 60% reduction in training time was reported for the proposed approach in comparison to the conventional SVM classifier. In terms of accuracy, the proposed approach was capable of predicting at risk and marginal students with up to 93.8% and 93.5% accuracy, respectively. However, the limitations of the presented study were not highlighted. Later, He et al. [9] proposed a novel Recurrent Neural Network-Gated Recurrent Unit (RNN-GRU) joint neural network to predict the at-risk students on VLE using both static (i.e., demographic) and sequential (i.e., engagement) data. Three RNN algorithms, including simple RNN, GRU and LSTM, were deployed on OULAD and reported that a simpler model achieved better performance in comparison to the complex LSTM model. The proposed approach was able to achieve 80% prediction accuracy. However, no comparison to existing literature was presented, and also the limitations of the presented were not highlighted. In the same year, Karimi et al. [15] developed a deep learning model for the prediction of students' performance on VLE using the demographic and engagement features from OULAD. Clickstream data was incorporated using the LSTM model, while demographic data was one hot encoded (i.e., a process in machine learning to convert categorical variables for model training) and feed to Fully Connected Network (FCN). From the results, for binary classification, the proposed model achieved an F1 score of up to 85%, while for the four-class classification, the proposed model was able to achieve an F1 score of up to 57%. Performance was compared with baseline SVM and LR models and was reported improvement for the proposed approach. Mubarak et al. [19] addressed the at-risk students prediction in the VLE problem as a time-series prediction. LR and Input-Output Hidden Markov Model (IOHMM) were implemented on students activity log data from OULAD to predict the early dropout. Performance of models was assessed using F1 score, accuracy and ROC. From the results, both the Sequential Logistic Regression (SELOR) and IOHMM achieved 84% accuracy and were best when compared to baseline machine learning models (i.e., LR, SVM, Decision Tree (DT), RF). However, no comparison with existing literature was presented to highlight the scope of this study.

Nazif et al. [20], in the year 2020, proposed the novel Probabilistic Neural Network (PNN) model with various feature selection algorithms (i.e., Minimum Redundancy Maximum Relevance (MRMR), Feature selection using neighborhood component analysis for classification (FSCNCA), Rank importance of predictors using ReliefF algorithm (ReliefF), Predictor importance estimates by permutation of out- of-bag predictor observations for random forest of regression trees (oobPermutedPredictorImportance), Infinite Feature Selection (infFS), Eigenvector Centrality Feature Selection (ECFS), Infinite Latent Feature Selection (ILFS), Mutual information Feature Selection (mutInfFS)) for the classification of students' performance on VLE using OULAD. The performance of the proposed approach was compared with the classical machine learning models (i.e., SVM, DT, LR, k-NN, RF, NB, Linear Discriminant Analysis (LDA)) and reported improved results. From the results, the proposed PNN approach outperformed conventional machine learning models and achieved 93.4% accuracy when used with the FSCNCA feature selector. However, no comparison with existing literature was reported to highlight the scope of the presented study. In the same year, Tomasevic et al. [23] employed various supervised machine learning approaches (i.e., k-NN, SVM, ANN, DT, Bayesian Linear Regression (BLR), Regularized Linear Regression (RLR)) to predict the exam performance (i.e., dropout risk, final exam score) on online platforms. A part of OULAD was used to train the machine learning models using engagement, performance and demographic features. The performance of trained models was assessed using the F1 score and RMSE measures. From the results, ANN was reported best with an F1 score of 96.62% when trained using engagement and performance features. The influence of demographic features on the performance prediction was reported as minimal. Although the performance of ANN was reported high, however, no discussion on why certain models performed best or worst was presented. Furthermore, no comparison with literature was presented to highlight the scope of the presented study. Waheed et al. [26] in year 2020 predicted the at-risk students using various deep learning models deployed using OULAD. Demographic, clickstream and assessment performance dataseet features were used to train the models. Performance was compared with conventional ML and reported improved 93% accuracy for deep learning model. However, no comparison with existing literature was presented to highlight the scope of study.

Ahmad et al. [2] in the year 2021 compared the performance of ANN and RF machine learning models for predicting the performance of the student on VLE using demographic and assessment features. Feature engineering was performed to select appropriate features from OULAD, and resulting data was used to train the models. From the results, ANN performed the best with a prediction accuracy of 91%. However, no cross-validation was performed to report the generalised performance. Furthermore, no comparison with existing literature was performed to highlight the scope of the presented study.

In one of the most recent studies, Fatema et al. [4] performed a comparative study using various machine learning models (i.e., k-NN, SVC, ANN) to predict the performance of students on VLE using different feature combinations of OULAD. Data was preprocessed as numerical values, and classification was applied to predict the performance as pass or fail. From the experimental investigations, k-NN and ANN were reported best with 99% prediction accuracy. Although high accuracy was reported; however, no cross-validation was performed in this case to deal with overfitting. Furthermore, feature engineering was not implemented as part of this research; however, the authors reported feature scaling as an important factor in improved performance. In 2020, Hidalgo et al. [11] implemented deep learning and meta-learning to predict the student's performance on VLE from behavioural data (i.e., click streams). A custom dataset extracted on the pattern of OULAD was used for this investigation; however, from a different online course. ANOVA and RFECV approaches were used to get the optimised hyperparameters for the model. From the results, the autonomous meta-learning-based model was able to achieve an accuracy of 67.2% on the test dataset that was comparable to the case of manual parameters and features selected/designed model.

Table 1: Summary of Presented Reivew of Benchmark Studies using OULAD to Assess Performance on VLE – Part A.

Heuer and Breiter	**Year**	**Entry Type**	**Technology Used**	**Dataset Features**	X	
Author	**Year**	**Entry Type**	**Technology Used**	**Dataset Features**	**Tasks**	**Results/Remarks**
Hlosta et al. [13]	2017	Conference	Ouroboros, Classical ML	Engagement and Demographic	At-risk students prediction	–XGBoost model was reported best for prediction when data from the only ongoing course was used.
Alshabandar et al. [5]	2018	Conference	GMM, k-NN and LR	Engagement and Demographic	Early dropout prediction	–GMM achieved an F1 score of 0.835 and reported best. –Latent engagement features proved effective in prediction.
Heuer and Breiter [10]	2018	Conference	Classical ML	Engagement, Demographic and Performance	Final exam success prediction	–SVM performed best with 87% accuracy. –Binarised daily activity has a similar impact as exact clickstreams on performance prediction
Aljohani et al. [3]	2019	Journal	LSTM and Conventional ML	Engagement (clickstream)	Early dropout and final success	–LSTM outperformed baseline models with 93.46% accuracy. –Student's final success can be predicted with high accuracy as early as ten weeks.
Jha et al. [14]	2019	Conference	Various Ensemble ML	Engagement, Demographic and Performance	Early dropout and final success	–GBM approach reported best with AUC and of 0.91 0.93 for dropout and final result prediction, respectively.
Chui et al. [6]	2020	Journal	RTV-SVM	Engagement, Performance and Demographic	At-risk/marginal students prediction	–RTV-SVM achieved approximately 94% prediction accuracy. –RTV-SVM reduced the training time by up to 59%.
He et al. [9]	2020	Journal	RNN-GRU	Engagement and Demographic	At-risk students prediction	–RNN-GRU achieved around 80% prediction accuracy. –Simpler RNN algorithms performed better than complex algorithms.
Karimi et al. [15]	2020	Conference	Deep Learning, SVM and LR	Engagement and Demographic	Performance prediction	–The proposed deep learning model achieved up to 85% for binary classification and 57% for four-class classification.
Mubarak et al. [19]	2020	Journal	SELOR and IOHMM	Engagement	At-risk students prediction	–SELOR and IOHMM achieved an accuracy of 84%. – Proposed models outperformed the baseline machine, learning models.

Most recently, in 2021, Hlioui et al. [12] proposed the use of behavioural indicators for the prediction of at-risk students on VLE using machine learning approaches. Data from OULAD was pre-processed and discretised using K-means before training the machine learning models. Furthermore, behavioural indicators were extracted, including autonomy, preservance, commitment, and motivation indicators were defined from the existing data. Various machine learning models (i.e., DT, RF, Bayesian, SVM, Multilayer Perceptron (MLP)) were trained using the unbalanced and balanced dataset with and without defined indicators to compare the performance. Models were assessed using an F1 score, and RF was reported best with an accuracy of 85% using a balanced dataset with discrete behavioural indicators. However, no cross-validation and comparison with existing literature was presented. Waheed et al. [25] in 2021 proposed the use of Sequential Conditional Generative Adversarial Network (SC-GAN) to address the class imbalance problem in the prediction of student's performance on VLE. The proposed SC-GAN generated synthetic data based on the previous records to solve the class imbalance and reported a 7% improvement in comparison to the conventional up-sampling approach of data balancing. As a classifier, RF and fully connected DNN were trained using the dataset. In the same year, Yuan et al. [27] investigated the detection of at-risk students in online courses based on their behavioural data. Nine classical machine learning models (i.e., LR, NB, RF, k-NN, SVM, Iterative Dischotomiser3 (ID3), Classification and Regression Trees (CART), Neural Network (NN), LSTM) were implemented on the pre-processed behavioural data (aggregate and sequence data) from OULAD. From the experimental results, RF and CART were reported best for the at-risk prediction using aggregate

Table 2: Summary of Presented Reivew of Benchmark Studies using OULAD to Assess Performance on VLE – Part B.

Heuer and Breiter	**Year**	**Entry Type**	**Technology Used**	**Dataset Features**	X	
Author	**Year**	**Entry Type**	**Technology Used**	**Dataset Features**	**Tasks**	**Results/Remarks**
Nazif et al. [20]	2020	Conference	PNN and Conventional ML	Engagement, Demographic and Performance	Final success prediction	–PNN with FSCNCA feature selection achieved the best accuracy of 93.4%.
Tomasevic et al. [23]	2020	Journal	Classical ML	Engagement, Demographic and Performance	At-risk students and final success	–ANN performed best with an F1 score of 96% using engagement and performance. –No impact of demographic features on the final exam score prediction
Waheed et al. [26]	2020	Journal	Deep Learning	Engagement, Demographic and Performance	At-risk students prediction	–Deep learning model outperformed conventional machine learning models with accuracy of 93%
Ahmad et al. [2]	2021	Conference	ANN and RF	Engagement, Demographic and Performance	Performance prediction	–ANN performed best with an accuracy of 91%. –Feature engineering was applied to achieve improved performance.
Fatema et al. [4]	2021	Conference	k-NN, SVC and ANN	Engagement, Demographic and Performance	Final success Prediction	–k-NN and ANN achieved a classification accuracy of 99%. –Improved results are potentially due to feature scaling.
Hidalgo et al. [11]	2021	Journal	ANOVA, RFECV and Meta-Learning	Engagement	Students performance prediction	–The meta-Learning approach was able to achieve 67.2% accuracy on the test dataset.
Hlioui et al. [12]	2021	Journal	Various Conventional ML	Engagement, Demographic and Performance	At-risk students prediction	–RF performed best with 84% accuracy using a balanced dataset with discrete behavioural indicators.
Waheed et al. [25]	2021	Journal	SC-GAN	Engagement, Demographic and Performance	At-risk students prediction	–SC-GAN achieved 7% improvement in comparison to conventional up-sampling approaches for data balancing.
Yuan et al. [27]	2021	Book Chapter	Classical ML and LSTM	Behavioural	At-risk students prediction	–SVM and CART for aggregate data LSTM for sequence data.

data, while the LSTM model was reported best for the sequence data. The performance of models was assessed by AUC value. Although encouraging results were reported, however, no comparison with existing literature was presented to demonstrate the effectiveness of the proposed investigation. Furthermore, potential limitations hindering the practical implementation of machine learning-based models were not reported. No insights about why certain models performed better or worst were reported from the experimental investigation.

As a summary, the presented review reported that OULAD had been used by researchers mainly to detect the at-risk students, detect the early dropout and predict the final exam performance. Various combinations of engagement, demographic and performance features were used in different scopes to predict the student's performance. Over the years, various learning-based approaches are deployed, including GMM, k-NN, LR, SVM, NB, RF and CART. The trend has been shifted from investigating the existing ML models towards developing new problem-specific learning-based models (e.g., RTV-SVM, IOHMM, SC-GAN, RNN-GRU). Furthermore, feature engineering and dealing with imbalanced data are two highlighted active areas of research where the state-of-the-art approaches have been implemented. Comparison with existing literature to highlight the scope of the presented study and the explanation of why certain models performed best or worst was found consistently lacking in reported studies.

5 CHALLENGES AND FUTURE DIRECTIONS

The presented review reported various learning-based models with high accuracies; however, there are still potential hinders and challenges causing the delay in real-world deployment of such algorithms. Followings are highlighted challenges and potential future work direction:

Feature Selection: Selection of appropriate features for the prediction of students' performance on VLE has been reported as one of the significant factors. In reported literature, this aspect has been only addressed by a couple of researchers and have huge potential for further exploration. Developing an optimal feature selection algorithm is the ultimate goal in this context.

State of the Art Approaches: Although reported literature implemented various learning-based advanced approaches; however, few approaches were found completely missing, such as encoder-decoder, transformers, 1-D CNN and ensemble modelling.

Feature Scaling: One major problem reported from the literature was significantly varied performances of the same models by different researchers. Because no author compared their results with existing literature, the reason for the varied performance was unknown. However, in this review, authors have remarked that each study has followed a different data pre-processing and scaling approaches, and that might be the reason for varied performance. A detailed study investigating the impact of different pre-processing techniques and feature scaling approaches is very much in the scope of this domain.

6 CONCLUSION

A review of benchmark research using the OULAD dataset for the performance assessment of students on VLE has been successfully presented. A range of learning-based algorithms have been reported in the literature to address the problem of students' efficient performance assessment. From the presented review, GMM, k-NN, LR, SVM, NB, RF, XGBoost, LR, ANN, LSTM, DRF, GBM, GLM, RTV-SVM, RNN-GRU, IOHMM, DT, PNN, LDA, BLR, RLR, ANOVA, RFECV, MLP, SC-GAN, ID3, and CART were reported as the highlighted technologies deployed in this regard. Furthermore, the OULAD dataset was reported to detect the at-risk students, detect the early dropout and predict the final exam performance. Limitations and challenges hindering the real-world implementation of learning-based models was found lacking in the literature. Feature engineering, deploying state of the art algorithms, feature scaling and dealing with class imbalance were highlighted as potential future research directions.

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